

06 — Deliverables and Milestones

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

1. Overview

This section outlines the concrete deliverables and milestones for the TriadicFrameworks project over the 12-month performance period. All deliverables will be openly licensed, publicly accessible, versioned, and archived with DOIs to ensure long-term reproducibility and community benefit.

The project is structured into four phases, each with clearly defined outputs and measurable milestones.

2. Deliverables

Deliverable 1 — dsrsp/0.1 Specification and Schemas

- finalized protocol specification
- JSON schemas for input envelopes and regime profiles
- RSM and vST alignment layer documentation
- example input/output files

Deliverable 2 — Reference Implementations

- open-source dsrsp/0.1 library
- RSM structural envelope generator
- vST validation block generator
- ILP module reference implementation

Deliverable 3 — Integration and Testing Materials

- synthetic datasets for testing
- reproducible Jupyter notebooks
- simulation-based integration examples
- validation workflows for RSM and vST

Deliverable 4 — Documentation and Developer Guides

- developer documentation for all components
- engine integration guide
- onboarding materials for contributors
- public website or documentation portal

Deliverable 5 — Community Release and Archival

- versioned public release of all components
- Zenodo-archived DOIs for specifications, code, and datasets
- virtual workshop or tutorial session
- final project summary and dissemination materials

3. Milestones by Project Phase

Phase 1 — Protocol Finalization (Months 1–3)

Milestones:

- M1.1: dsrsp/0.1 draft specification completed
- M1.2: RSM and vST alignment layer defined
- M1.3: schemas and example files published
- M1.4: initial Zenodo archival of protocol components

Phase 2 — Reference Implementations (Months 3–6)

Milestones:

- M2.1: dsrsp/0.1 library implemented
- M2.2: RSM structural envelope generator completed
- M2.3: vST validation block generator completed
- M2.4: ILP module implemented and tested

Phase 3 — Integration and Testing (Months 6–9)

Milestones:

- M3.1: dsrsp/0.1 integrated into simulation environment
- M3.2: regime classification workflows validated
- M3.3: synthetic datasets and notebooks published
- M3.4: cross-component integration tests completed

Phase 4 — Documentation and Community Release (Months 9–12)

Milestones:

- M4.1: developer documentation completed
- M4.2: integration guides published
- M4.3: public release of TriadicFrameworks v0.1
- M4.4: virtual workshop delivered
- M4.5: final archival of all deliverables with DOIs

4. Completion Criteria

The project will be considered complete when:

- all deliverables listed above are publicly available,
 - all components are versioned and archived with DOIs,
 - documentation and integration guides are published,
 - reference implementations are functional and tested,
 - a community workshop or tutorial has been delivered, and
 - a final project summary has been submitted to NASA.
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5. Summary

The deliverables and milestones outlined in this section ensure that TriadicFrameworks will provide a coherent, reproducible, and openly accessible substrate for regime-aware scientific analysis. The phased structure supports steady progress, transparent development, and a high-quality public release aligned with NASA's open-science mission.